MILITARY SPECIFICATION

RADIACMETER IM-174( )/PD

This specification is approved for use by the Electronics Research and Development Command, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a portable survey meter that detects and measures gamma radiation, and is designated Radiacmeter IM-174( )/PD (see 6.5).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-B-18	- Battery, Dry.
MIL-P-116	- Preservation, Methods Of.
MIL-P-11268	- Parts, Materials, And Processes Used In Electronic Equipment.
MIL-M-13231	- Marking Of Electronic Items.
MIL-F-14072	- Finishes For Ground Electronic Equipment.
MIL-E-55585	- Electronics Equipment And Parts, Packaging Of.

STANDARDS

MILITARY

MIL-STD-105	- Sample Procedures And Tables For Inspection By Attributes.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Electronics Research and Development Command, ATTN: DELCS-PE, Fort Monmouth, New Jersey 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6665

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MIL-STD-252	- Wired Equipment Classification of Visual And Mechanical Defects.
MIL-STD-454	- Standard General Requirements For Electronic Equipment.
MIL-STD-726	- Packaging Requirements Code.
MIL-STD-810	- Environmental Test Methods.

2.1.2 Other Government documents. The following Government drawings form a part of this specification to the extent specified herein:

DRAWINGS

ELECTRONICS COMMAND

DL-SM-B-537154	- Radiacmeter IM-174( )/PD
DL-SM-B-537155	- Case, CY-2446( )/PD

(Copies of specifications, standards, and drawings required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The Radiacmeter IM-174( )/PD is a portable, unsaturated ionization chamber type survey meter operated from batteries contained within the instrument. The meter detects and measures gamma radiation. Radiation intensity is indicated on a pseudo-logarithmic scale. A spring loaded lever action switch provides facilities for zero setting in a radiation field and checking the circuit of the amplifier.

3.2 First article. The quantity of samples specified in the contract shall be subjected to first article inspection (see 4.3 and 6.2b).

3.3 Material. In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirement therein shall apply. The material shall be as shown on the drawings and as specified herein. However, when a definite material is not specified, a material shall be used which will enable the radiacmeter to meet the requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.3.1 Recycled material. It is encouraged that recycled material be used, when practical, provided it meets the requirement of this specification.

3.4 Design. The radiacmeter and case shall be in accordance with DL-SM-B-537154, DL-SM-B-537155, and as specified herein.

3.4.1 Ion chamber assembly.

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3.4.1.1 Wall coating. The material used to form the conducting wall of the ionization chamber shall be graphite alcohol dag. The chamber wall shall be cleansed of parting compound prior to coating. The coating shall be opaque, and uniform with no flaking, peeling, or alligatoring.

3.4.1.2 Assemblage. The ionization chamber and the grease-filled compartment containing the high impedance circuitry shall be cast in a monolithic block of epoxy resin. A connector plug, Continental Connector Co. #8-20P or equal, shall be cast into the monolithic block to provide for all external connections. Prior to final sealing, the assembly shall be placed in an enclosure capable of being evacuated and subjected to vacuum of at least 1.3 inches of mercury (in.Hg.) (4.4kPa) for 15 minutes and shall then be visually examined. No grease, sealant or other materials should be allowed to leak from one compartment to another within the assemblage. After assembly, the ionization chamber shall be a hermetically sealed unit.

3.4.2 Power supply. The radiacmeter shall operate from a battery power supply contained in a watertight compartment within the equipment. Unless otherwise specified (see 6.2c), batteries shall not be provided for shipment.

3.4.2.1 Compartment. The battery compartment shall not compromise the watertight seal to the interior of the radiacmeter.

3.4.2.2 Batteries. The power supply shall consist of batteries conforming to MIL-B-18 as follows:

BA-30 (For use at or above -12°C (10°F))
BA-3030 (For use below -12°C (10°F))

3.5 Operation. When tested as specified in 4.6.2 the radiacmeter shall demonstrate the following conditions:

a. Continuous operation for a period of one year, alternating 10 hours on and and two hours off, with no more than normal replacement of batteries. Aceptance of rist article samples will satisfy this requirement.

b. Continuous operation at an ambient temperature between -18°C (0°F) and 0°C (32°F) for a period of 72 hours in ambient background radiation. Battery replacement will not be permitted during this test.

3.6 Performance.

3.6.1 Calibration. All calibration checks shall be performed with a Cobalt 60 or Cesium 137 source which has been calibrated, standardized and certified by the National Bureau of Standards within 6 months prior to start of the preproduction tests.

3.6.2 Sensitivity and accuracy. The range of the radiacmeter shall be from 1 to 500 rads/hr. The radiacmeter shall indicate zero when located in a radiation field below 100 millirads/hr. The accuracy shall be ±15 percent.

3.6.3 Time constant. The radiacmeter after a 20 minute warm-up, shall indicate not less than 90 percent of its total response to a known applied radiation rate within 3 seconds after exposure. When the applied radiation rate is removed the radiacmeter shall indicate not more than 10 percent of its total response within 3 seconds after removal.

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3.6.4 Air pressure sensitity. The radicmeter shall not deviate from its indicated dose rate at sea level (30.0 in.Hg. (101kPa)) when operated under reduced pressured of 20.58 in.Hg. (69.5kPa), and subjected to 24 rads/hr ±40 percent for a period of 24 hours.

3.6.4.1 Modified air pressure sensitivity. Identical to that specified in 3.6.4 except that the duration shall be 5 minutes in lieu of 24 hours.

3.6.5 Drift and stability.

3.6.5.1 Drift. Short term stability shall be such that, after a warm-up period of 20 minutes, readjustment to maintain accuracy of the radiacmeter to within ±15 percent of true dose rate shall not be required during a continuous period of not less than 6 hours at any temperature between 10°C (50°F) and 27°C (81°F).

3.6.5.2 Stability. When tested as specified in 4.6.3.4.2, the radiacmeter shall retain its calibration accuracy within ±15 percent of the initial reading for a period of not less than 4 months. Batteries may be replaced as required; battery replacement data must be recorded. No adjustment of the calibration potentiometer is permitted during the period of this test.

3.6.6 Warm-up. The radiacmeter shall respond, with an accuracy of ±20 percent of true dose rate, within 5 minutes after switching on the instrument. The required calibration accuracy in accordance with 3.6.8 shall be obtainable within 20 minutes after the radiacmeter is switched on.

3.6.7 Energy dependence. The response of each radiacmeter shall be as uniform as possible when exposed to equal rad doses of X-ray and gamma radiation whose energies lie between 80 thousand electron volts (keV) and 3.0 million electron volts (MeV). That is, when each radiacmeter is irradiated with equal rad doses of X or gamma rays of energies in its specified range, and the response of the radiacmeter to each of these doses is normalized to the response of the radiacmeter to the same dose of Cobalt 60 or Cesium 137 radiation, the normalized values shall be as close to unity as possible and shall in no case lie outside the range of 0.8 to 1.2 ±20 percent. For energies lower than 80 keV, the response of each radiacmeter shall, as far as is practicable, match that of the human body, and in all cases exhibit a continual decrease in sensitivity for equal doses of lower energy until the response diminishes to not more than 5 percent at 20 keV; (extrapolation down to 20 keV is acceptable, if it is a continuation of a natural curve and at least one point lies below 50 keV).

3.6.8 Calibration accuracy. The radiacmeter shall be calibrated at standard temperature, 23° ±10°C (73° ±18°F), over its entire scale from a standardized Cobalt 60 or Cesium 137 source to within ±15 percent of true dose rate. Once calibrated, the accuracy of any scale reading shall be within ±15 percent of the true dose rate at any temperature between -32°C (-25°F) and +53°C (+127°F) as measured with the instrument specified in 4.1.1b.

3.6.9 Indicator fall-off. When the radiacmeter is exposed to radiation intensities in excess of 500 rads/hr, it shall indicate at least a full scale reading and at no time indicate a decreasing response as the dose rate is increased.

3.6.10 Compatibility. Each properly operating radiacmeter shall indicate 10 rads/hr ±15 percent and 100 rads/hr ±15 percent when checked with radiac calibrator set AN/UDM-2. A properly operating radiacmeter shall be defined as one meeting the requirements specified in 3.6.8.

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3.6.11 Controls

3.6.11.1 Potentiometer setting. After adjustment of the calibration and zero-set potentiometers for proper instrument operation, the radiacmeter shall operate as specified in 3.6.11.1.1 and 3.6.11.1.2.

3.6.11.1.1 Calibration control. In a properly operating radiacmeter, the control rotor shall be within +75° or -110° of the center of the dynamic range of the control. Rotation in a counter clockwise (CCW) direction shall be designated by a negative (-) sign while rotation in a clockwise (CW) direction shall have a positive (+) sign.

3.6.11.1.2 Zero-set control. In a properly operating radiacmeter, the control rotor shall be at the center of its dynamic range ±110° of rotation.

3.6.12 Performance check. When tested as specified in 4.6.4 the radiacmeter shall retain its calibration accuracy within ±15 percent of true field.

3.6.13 Microphonics. The radiacmeter shall exhibit no microphonic effects.

3.7 Environmental conditions. The radiacmeter shall withstand individual environmental conditions specified herein, or combinations thereof, without damage, performance degradation, or malfunction when tested as specified in 4.6.6.

3.7.1 Operating.
a. Temperature range: -32°C (-25°F) to 52°C (125°F).
b. Humidity: Up to 98 percent relative humidity.
c. Altitude: Up to 10,000 feet (3048m).

3.7.2 Nonoperating (storage).
a. Temperature range: -51°C (-60°F) to 71°C (160°F).
b. Humidity: Up to 98 percent relative humidity.
c. Altitude: Up to 50,000 feet (15.24km).

3.7.3 Immersion. The radiacmeter shall show no evidence of water in either the battery compartment or instrument case when immersed in water to a depth of 36 inches (0.9m).

3.7.4 Salt fog. When tested as specified in 4.6.6.6 the radiacmeter shall show no evidence of degradation in performance or exfoliation of metal, such as flaking, pitting, blistering or loosening of finish on metal surfaces and there shall be no clogging or binding of moving parts.

3.7.5 Dust (fine sand). When tested as specified in 4.6.6.7 the radiacmeter shall withstand the effects of blowing dust particles and shall be resistant to particles that may accumulate within the assembly without showing evidence of degradation in performance.

3.7.6 Vibration, bounce and shock.

3.7.6.1 Vibration. When tested as specified in 4.6.6.8.1 the radiacmeter shall show no evidence of degradation in performance or deviation of air pressure sensitivity.

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3.7.6.2 Bounce. When tested as specified in 4.6.6.8.2 the radiacmeter shall show no evidence of degradation in performance or physical damage beyond minor surface abrasions.

3.7.6.3 Shock (drop). The radiacmeter shall show no degradation in performance and only minor physical damage when dropped from a height of 4 feet (1.2m).

3.7.7 Fungus. When tested as specified in 4.6.6.8.4 the radiacmeter shall reveal no evidence of viable fungus on the surface.

3.8 Bounce preconditioning. When test as specified in 4.6.7 the radiacmeter, without subsequent processing, shall show no evidence of degradation in performance.

3.9 Interchangeability. Like units, subassemblies, and replaceable parts shall conform to requirement 7 of MIL-STD-454.

3.10 Weight. The weight of the radiacmeter with batteries installed, shall be not more than 66 ounces (1.9kg).

3.11 Finish. The equipment shall be finished in accordance with MIL-F-14072 and the equipment drawings. This includes finish of hardware, such as handles, hinges, screws, etc., and necessary touch-up after mounting. The final paint film on type 1 surfaces shall be forest green.

3.12 Marking. Marking shall conform to MIL-M-13231.

3.12.1 Visibility. Whenever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the radiacmeter.

3.12.2 Serial numbers. Each radiacmeter shall bear a serial number in accordance with MIL-M-13231.

3.13 Government-loaned property. Radiac calibrator set AN/UDM-2, NSN 6665-00-179-9037 shall be furnished by the Government on a loan basis for testing purposes (see 6.4).

3.14 Workmanship. The radiacmeter shall be manufactured and assembled in accordance with requirement 9 of MIL-STD-454.

4. QUALILTY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract , the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract , the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Contractor's special facilities. The contractor shall be required to provide in his plant, or have direct access to the following facilities:

a. Cobalt 60 or Cesium 137 source capable of providing a minimum field intensity of 125 rads/hr at a distance of 1 meter (3.3 ft.) from its geometrical center
b. A Victoreen Condenser "R" meter, or equivalent, calibrated by the National Bureau of Standards or other agency approved by this command for measuring radiation flux.
c. A calibrated source of X-rays or low energy gamma rays suitable for use in the determination of the energy dependence of the ionization chambers.
d. Climatic test equipment.
e. Durability test equipment.

(To comply with the climatic requirements of this specification the facilities listed in a and d above must be located in the same immediate area.)

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Inspections covered by subsidiary documents (see 4.4).
c. Quality conformance inspections (see 4.5).
d. Packaging inspection (see 4.7).

4.3 First article.

4.3.1 First article units. When specified in the contract, the contractor shall furnish first article units of the complete radiacmeter.

4.3.2 First article inspection. The first article inspection shall consist of the inspections specified in table I and shall be performed in the order specified. Failure of any inspection shall be cause for rejection.

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TABLE I. First article inspection.

Inspections	Requir para	Test para	Units 1-5	Units 6-10	Units 11-15	Units 16-19	Units 20-24	Units 25-29	Units 30-31
1. Inspections covered by 1/ subsidiary documents	3.11,3.12 and 3.14	4.4	Inspection to be performed on all units.						
2. Bounce preconditioning 1/	3.8	4.6.7	Inspection to be performed on all units.						
3. Weight 1/	3.10	4.6.9	Inspection to be performed on all units.						
4. Group A inspection 1/	See table II		Inspection to be performed on all units.						
5. Group B inspection 1/	See table III		Inspection to be performed on all units.						
6. Carrying case with strap	3.14	4.6.1	Inspection to be performed on five carrying cases with straps.						
7. Environmental 2/									
High Temperature	3.7.1a and 3.7.2a	4.6.6.1	1						
Low Temperature	3.7.1a and 3.7.2a	4.6.6.2	2						
Vibration	3.7.6.1	4.6.6.8.1		1					
Bounce	3.7.6.2	4.6.6.8.2		2					
Temperature-altitude	3.7.1c and 3.7.2c	4.6.6.4			1				
Humidity	3.7.1b and 3.7.2b	4.6.6.3			3				
Shock (drop)	3.7.6.3	4.5.5.8.3				1			
Salt fog 3/	3.7.4	4.6.6.6				2			
Dust	3.7.5	4.6.6.7					2		
Immersion	3.7.3	4.6.6.5			2				
Energy dependence	3.6.7	4.6.3.6					1		
Operation	3.6.12	4.6.2	3						
Stability	3.6.5.2	4.6.3.4.2						1	
Fungus 3/	3.7.7	4.6.6.8.4							1

1/ The inspections 1 through 5, in the order shown, shall be performed on all first article units before subjecting these units to any other inspection.

2/ The numbers in the units columns in the table covered by inspection item 7 specify the order of inspections for the indicated units.

3/ The radiacmeter shall be thoroughly washed, cleaned, dried and refurbished after the salt fog and fungus tests before proceeding with subsequent inspection.

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4.4 Inspections covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification, and the inspection requirement specified in the contract :

Item	Where required
Finish	3.11
Marking	3.12
Workmanship	3.14

4.5 Quality conformance inspection. The contractor shall perform the inspections specified in 4.4, 4.5.1 through 4.5.3, 4.6.7 and 4.6.8. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection. Each unit shall be inspected for conformance to the inspections specified in table II. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels, and acceptable quality levels (AQLs) indicated in table II.

4.5.1.1 Order of inspection within group A. Group A inspection shall be performed in an order satisfactory to the Government.

TABLE II. Group A inspection.

Inspection	Requirement paragraph	Test paragraph	AQL (percent defective) Major	AQL (percent defective) Minor
Visual and mechanical				
Ionization chamber (complete assembly)	3.4.1	4.6.1	0.65	2.5
Radiacmeter (complete w/o carrying case)	3.14	4.6.1	1.0	4.0
Carrying case with strap	3.14	4.6.1	1.0	4.0
Electrical				1/
Calibration accuracy	3.6.8	4.6.3.7		
Warm-up	3.6.6	4.6.3.5	1.5 for	
Time constant	3.6.3	4.6.3.2	the group	
Potentiometer setting	3.6.11.1	4.6.3.10		
Microphonics	3.6.13	4.6.5	1	
Sensitivity and calibration	3.6.2	4.6.3.1	1	

1/ All electrical defects are major.

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4.5.2 Group B inspection. Group B inspection shall normally be performed in inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection. This inspection shall conform to table III and to the special inspection levels of table I of MIL-STD-105.

4.5.2.1 Group B sampling plans. The group B sampling plans, for the AQLs listed in table III shall be as follows:

AQL	Inspection level
2.5 percent	S-3
4 percent	S-4

4.5.2.2 Order of inspection within group B. Group B inspection shall be performed in an order satisfactory to the Government.

TABLE III. Group B inspection.

Inspection	Requirement paragraph	Test method paragraph	AQL (percent defective)
Interchangeability	3.9	4.6.8	4
Compatibility	3.6.10	4.6.3.9	4
Air pressure sensitivity	3.6.4	4.6.3.3	
Drift	3.6.5.1	4.6.3.4.1	2.5 for
Indicator fall-off	3.6.9	4.6.3.8	the group

4.5.3 Group C inspection. Group C inspection shall be performed on units that have passed group A and group B inspection. The inspection shall consist of the inspections specified in table IV.

4.5.3.1 Sampling for group C inspection. Thirteen samples of the complete radiacmeter shall be selected at random, 4 samples for each subgroup 1 through subgroup 3, and one sample for subgroup 4 in table IV. The samples selected shall be at the start of the contract from the first quality conformance inspection production lot. The fungus test shall be performed only on the first production lot. Thereafter, 12 samples of the complete radiacmeter shall be selected at random, 4 samples for each subgroup 1 through subroup 3 in table IV. The samples selected shall be at the 25 percent, 50 percent and 75 percent points in production ($\pm$ 5 days).

4.5.3.2 Order of inspection within group C. The order of the tests may be varied within each subgroups to suit the availability of test facilities and decrease total testing time.

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TABLE IV. Group C inspection.

Inspection	Requirement paragraph	Test method paragraph
Subgroup 1		
Vibration	3.7.6.1	4.6.6.8.1
Temperature-altitude	3.7.1c, 3.7.2c	4.6.6.4
Engery dependence	3.6.7	4.6.3.6
Immersion	3.7.3	4.6.6.5
Subgroup 2		
Bounce	3.7.6.2	4.6.6.8.2
Humidity	3.7.1b, 3.7.2b	4.6.6.3
High temperature	3.7.1a, 3.7.2a	4.6.6.1
Low temperature	3.7.1a, 3.7.2a	4.6.6.2
Subgroup 3		
Dust 1/	3.7.5	4.6.6.7
Salt fog 1/	3.7.4	4.6.6.6
Shock (drop)	3.7.6.3	4.6.6.8.3
Operation	3.6.12	4.6.2
Subgroup 4		
Fungus 1/	3.7.7	4.6.6.8.4

1/ The radiacmeter shall be thoroughly washed, cleaned, dried and refurbished after their inspection before proceeding with subsequent inspection.

4.5.3.3 Group C failures. Actions required relative to group C failures shall be as specified in the contract (see 6.2).

4.6 Methods of inspection.

4.6.1 Visual and mechanical inspection. The radiacmeter shall be examined for the applicable defects listed in MIL-STD-252. The carrying case and straps shall be examined for workmanship, fit and application of specified material treatment. The five units from the first article phase shall be used to determine the fit of the carrying case. Classification of visual and mechanical defects for the carrying case and strap shall be as shown below:

Classification	Defect
Major	End clips missing or loose. Broken stitches. Fastening devices: -Missing. -Inoperable. -Incorrectly attached.

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	Loose stitching. Holes or cuts in material. Fit, dimension small, will not fit sample. Strap dimensionally short.
Minor	Color, slightly off. Marking essentially as specified but not per print. Fit, dimension large. Strap, dimensionally, long. Stains on material.

4.6.2 Operation. Place the instrument and/or ion chamber in a jig or fixture capable of holding them and a radiation source so that the geometry of radiation is exactly reproducible. Install a radiation source, capable of providing a dose rate of 25 rads/hr ±40 percent, in the fixture. Install fresh bateries in the instrument and place jig or fixture, into a temperature chamber and allow the temperature to stabilize at -18°C (0°F). Turn on the instrument, allow it to warm up for a period of 20 minutes and check the performance of the "check-zero" switch in accordance with 4.6.4. With the instrument in its normal operating mode, record the meter reading. Leave the instrument operating for a period of 72 hours; the source may be removed or left in the jig throughout the 72-hour test at the discretion of the contractor. At the end of the 72-hour period check the performance of the "check-zero" switch (if source has been removed, reinstall) and with the instrument in its normal operating mode, record meter reading. Nonconformance to 3.5b shall constitute failure of this test.

4.6.3 Performance.

4.6.3.1 Sensitivity and calibration. A sensitivity and calibration test shall be performed on each instrument as follows:

a. The instrument shall be mounted in a jig in such a manner that the calibration geometry is reproducible for other instruments to be similarly tested.

b. The instrument shall be subjected to a beam from a radiation source and the incident radiation flux shall be determined by an approved meter.

c. The geometric center of the ionization chamber and the central axis of the central axis of the calibrating beam shall coincide.

d. The instrument shall be mounted in a vertical position with the calibration control facing upwards and the zero-set potentiometer knob facing into the beam. The beam shall be directed normal to the top broad face of the ionization chamber.

e. The instrument shall be turned on and allowed to warm up for 20 minutes prior to commencing calibration.

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f. The instrument shall be set to zero prior to calibration. The calibration control cover shall be removed and the control adjusted by means of a manipulator so that the indicator reads 100 rads/hr when the calibration rate is applied. The instrument shall return to zero indication upon removal of the calibration field.

g. Set the radiation field to 100 rads/hr and place the lever-action switch in the zero position. If the indicator fails to read zero, zero the instrument while it is in the radiation field. Set the lever-action switch to the normal position. The indicated reading shall be 100 rads/hr +15 percent.

h. The lever-action switch shall then be held in the "check" position while the instrument is in the calibrating beam. The indicator shall show a steady reading within the marked portion of the scale.

i. The instrument shall be placed in a true field of 10 rads/hr. The instrument reading shall be 10 rads/hr ± 15 percent.

4.6.3.1.1 Failure criteria. Nonconformance to 3.6.2 shall constitute failure of this test.

4.6.3.2 Time constant. Place instrument at a distance from the cobalt or cesium source such that the field at that location will be 50 rads/hr when the source is opened. Open source suddenly and time the interval for the meter needle to rise from zero to 90 percent of final reading (i.e. to 45 rads/hr). After a wait of approximately 30 seconds to allow complete stabilization at 50 rads/hr, close the source suddenly and time the interval for the meter needle to fall from 50 rads/hr to 10 percent of that reading (i.e. to 5 rads/hr). Nonconformance to 3.6.3 shall constitute failure of this test.

4.6.3.3 Air pressure sensitivity. Remove the cover assembly from the instrument, turn it on and allow the instrument to warm up for a period of 20 minutes. Adjust the zero setting and place the instrument in a vacuum chamber. Fix the position of a radiation source, capable of providing a dose rate of about 24 rads/hr, relative to the ionization chamber of the instrument and note the instrument reading. This initial reading shall be 24 rads/hr ± 40 percent. Decrease the chamber pressure to 20.58 in.Hg. (69.5kPa). Maintain this for a period of 24 hours. The instrument indication shall be recorded once every hour for the first 6 hours and at the end of the test. There shall be no drift at any time greater than ± 15 percent of the intitial reading. An apparatus permitting the operation of the detector chamber remote from the instrument may be used to carry out this test. Nonconformance to 3.6.4 shall constitute failure of this test.

4.6.3.3.1 Modified air pressure sensitivity. This test shall be identical to that of 4.6.3.3 except that the duration shall be 5 minutes in lieu of 24 hours. Noncormance to 3.6.4.1 shall constitute failure of this test.

4.6.3.4 Drift and stability.

4.6.3.4.1 Drift. Turn on instrument and allow a 20 minute warm-up period. Expose the instrument to a source which produces a meter reading between 15-35 rads/hr for a period of 6 hours. At the end of the 6-hour period the meter indication shall be compared with the initial reading. Nonconformance to 3.6.5.1 shall constitute failure of this test.

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4.6.3.4.2 Stability. Calibrate the instrument in accordance with 4.6.3.1 and measure the calibration accuracy specified in 4.6.3.7; record these initial readings. Subject the instrument to a duty cycle of 6 hours on, 18 hours off, 5 days per week for a total of 4 months. The calibration accuracy shall be measured once each week and compared to the initial readings. Nonconformance to 3.6.5.2 shall constitute failure of this test.

4.6.3.5 Warm-up. Turn on the instrument and after a 5 minute warm-up, expose it to a true does rate of 10 rads/hr. The instrument shall be within ±20 percent of the true dose. Repeat the above for a dose rate of 100 rads/hr. Nonconformance to 3.6.6 shall constitute failure of this test.

4.6.3.6 Energy dependence calibration.

4.6.3.6.1 Test equipment.

4.6.3.6.1.1 Potential. A constant potential X-ray machine operable up to a maximum voltage of 250 KV shall be employed for calibration tests in the low energy region.

4.6.3.6.1.2 Collimating systems. A diaphragming or collimating system shall be used to define the X-ray beam so that the instrument being tested will be irradiated uniformly and scattering from walls or neighboring objects will be minimized.

4.6.3.6.1.3 Effective wavelength. The data on X-ray tube potentials, filtration, energy at maximum intensity and the half height limits on energy spread as obtained by Kramer's rule are shown in table V. A Westinghouse quadrocondex 250 KVP constant potential machine or equal associated with an X-ray tube with an inherent filtration equal to 3 millimeters (mm) of aluminum is required.

TABLE V. X-ray tube data.

Tube applied potential (Kv)	Filtration (mm)	Bean energy at maximum intensity (KeV)	Energy spread at half heights (keV)	
40	1.67 AL	30	- 5.3	+8.0
70	0.89 Cu	56	-10.1	+10.05
100	1.07 Sn	85	-14.0	+10.5
150	4.28 Sn	129	-19.0	+16.3
200	2.00 Pb 2.14 Sn	178	-23.0	+15.5
250	400 Pb 2.14 Sn	222	- 28.0	+21.5

4.6.3.6.2 Test procedure. Expose the instrument to various dose rates over the specified range of the equipment using a Cobalt 60 or Cesium 137 source and record the readings. The instrument shall then be exposed to the same true dose rates as above; however, these rates shall have various energies over the range of 80 keV to 3.0 MeV. The true dose rate shall be determined by a Victoreen R-meter, that has been calibrated in accordance with 4.1.1b. Record the instrument's reading for each exposure. These values shall then be compared to those obtained from Cobalt 60 or Cesium 137 to determine compliance with 3.6.7. The instrument shall then be exposed to the same dose rates as above but the energies involved shall be below 80 keV. Nonconformance to 3.6.7 shall constitute failure of this test.

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4.6.3.7 Calibration accuracy. Turn on the instrument and after a 20-minute warm-up expose it to true dose rates of 3, 5, 10, 50, 100, 200, and 400 rads/hr and record readings. The test shall be performed at standard temperature. Nonconformance to 3.6.8 shall constitute failure of this test.

4.6.3.8 Indicator fall-off. Turn on equipment and allow a 20 minute warm-up period. Expose detector to fields from 500 rads/hr to 5000 rads/hr over a period of 5 minutes. Nonconformance to 3.6.9 shall constitute failure of this test.

4.6.3.9 Compatibility. The equipment shall be checked in Radiac Calibrator AN/UDM-2 in accordance with the established procedures contained in the calibrator's technical manual. Nonconformance to 3.6.10 shall constitute failure of this test.

4.6.3.10 Potentiometer settings. Turn on instrument and allow a 5-minute warm-up. With zero-check switch in zero position, zero the instrument; release switch. Check physical position of zero potentiometer rotor and calibration potentiometer rotor. Nonconformance to 3.6.11 shall constitute failure of this test.

4.6.4 Performance check. Turn on instrument and allow at least a 3-minute warm-up. With function switch in "zero" position adjust zero set control to zero. With function switch in "check" position, meter shall read in the "check-band;" release function switch and meter shall read zero. Expose instrument to true field of 100 rads/hr and 10 rads/hr. Press the bottom for the meter light; the lamp shall light. Nonconformance to 3.6.12 shall constitute failure of this test.

4.6.5 Microphonics. Each instrument shall be rested on its largest face on a level, solid surface. A block, measuring 2 inches (51mm) cube, with a length of twine attached, shall be inserted under the leading edge of the instrument. When the block is quickly removed by pulling sharply on the twine, a meter deflection caused by the resultant blow exceeding 25 percent of the total scale length shall constitute failure of this test.

4.6.6 Environmental.

4.6.6.1 High temperature. Place the instrument, switch on, in the test chamber. Raise the internal chamber temperature to 49°C (120°F) and maintain this condition for a minimum of 3 hours after instrument temperature stabilization. Raise the temperature to 53°C (127°F) for one hour. Do a performance check as specified in 4.6.4 during the last 30 minutes of this one hour period. Upon completion of the performance check, switch the instrument off and raise the chamber temperature to 71°C (160°F). Maintain at this temperature for a period of at least 4 hours after instrument temperature stabilization. Return the instrument to standard ambient conditions and stabilize.

4.6.6.2 Low temperature. Place the instrument, switch on, in the test chamber. Lower the internal chamber temperature to -32°C (-25°F) and after temperature stability has been reached, maintain this condition for at least 6 hours. At the end of this 6-hour period, do a performance check as specified in 4.6.4. Switch off the instrument and low chamber temperature to -51°C (-60°F) and hold at this temperature for 4 hours after the instrument temperature stabilized. Upon completion of this period, return the instrument to standard ambient conditions and stabilize.

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4.6.6.3 Humidity. The instrument shall be subjected to the test of Method 507.1, Procedure II of MIL-STD-810. Operation is not required during the humidity cycles.

4.6.6.4 Temperature-altitude. Remove the cover assembly from the instrument and position a radiation source capable of providing a dose rate of about 24 rads/hr ± 40 percent relative to the ionization chamber of the instrument. Place the instrument in a vacuum chamber and place the combination in a temperature chamber. Turn instrument on and after a 3-minute warm-up, record the instrument reading. Switch instrument off, lower the vacuum chamber pressure to 20.58 in.Hg. (69.5kPa), and the temperature chamber to -32°C (-25°F); maintain this condition for 12 hours. Upon completion of the 12 hour period, switch the instrument on after a 3-minute warm-up, record the reading and turn the instrument off. Lower the vacuum chamber pressure to 3.4 in.Hg. (11.5kPa), and the temperature chamber to -51°C (-60°F) and maintain for 4 hours. Upon completion of the 4-hour period return instrument to room conditions, turn the instrument on and after a 3-minute warm-up, record the reading. Compare readings obtained at start of test, at end of -32°C (-25°F) period and at the end of the test. An apparatus permitting the operation of the detector chamber remote from the instrument may be used to carry out this test.

4.6.6.5 Immersion. The instrument shall be subjected to the test of Method 512.1, Procedure I of MIL-STD-810 except that the test items and the water shall be at room temperature.

4.6.6.6 Salt fog. The instrument shall be subjected to the test of Method 509.1, Procedure I of MIL-STD-810. The applicable salt solution shall be 20 percent. The performance tests required before and after the subjection to the salt environment shall be as specified in 4.6.4.

4.6.6.7 Dust (fine sand). The instrument shall be subjected to the test of Method 510.1, Procedure I of MIL-STD-810. After testing, the instrument shall meet the performance check as specified in 4.6.4.

4.6.6.8 Vibration, bounce and shock.

4.6.6.8.1 Vibration. Clamp the instrument, with batteries installed, firmly to the bed of a vibration table adjusted to cycle between 10 and 55 cycles per second with a total excursion of 0.015 inches (0.38mm). Vibrate for 1 hour, stop and reclamp the instrument in a second plane at right angles to the first position. Vibrate for 1 hour, stop and test the instrument as specified in 4.6.3.3.1 and 4.6.4.

4.6.6.8.2 Bounce. The instrument shall be subjected to the test of Method 514.2, Procedure XI, Part 2 of MIL-STD-810. The inside surfaces of the wooden fences shall be lined with felt not more than 0.25 inch (6.4mm) thick. During each hour of the 3-hour test period, the instrument shall rest on a different face.

4.6.6.8.3 Shock (drop). The test shall be carried out on a surface of 2 inch (51mm) plywood backed by concrete. The sequence of drops to be applied to the instrument shall be as numbered in figure 1, i.e. 1-10. By following this sequence the drops are automatically paired to be diametrically opposite, i.e., 1-2, 3-4 etc. Two instruments shall be used to receive 10 drops, one instrument for drops 1 to 6 inclusive and one for drops 7 to 10 inclusive. The instrument shall be dropped from a height of 4 feet (1.2m).

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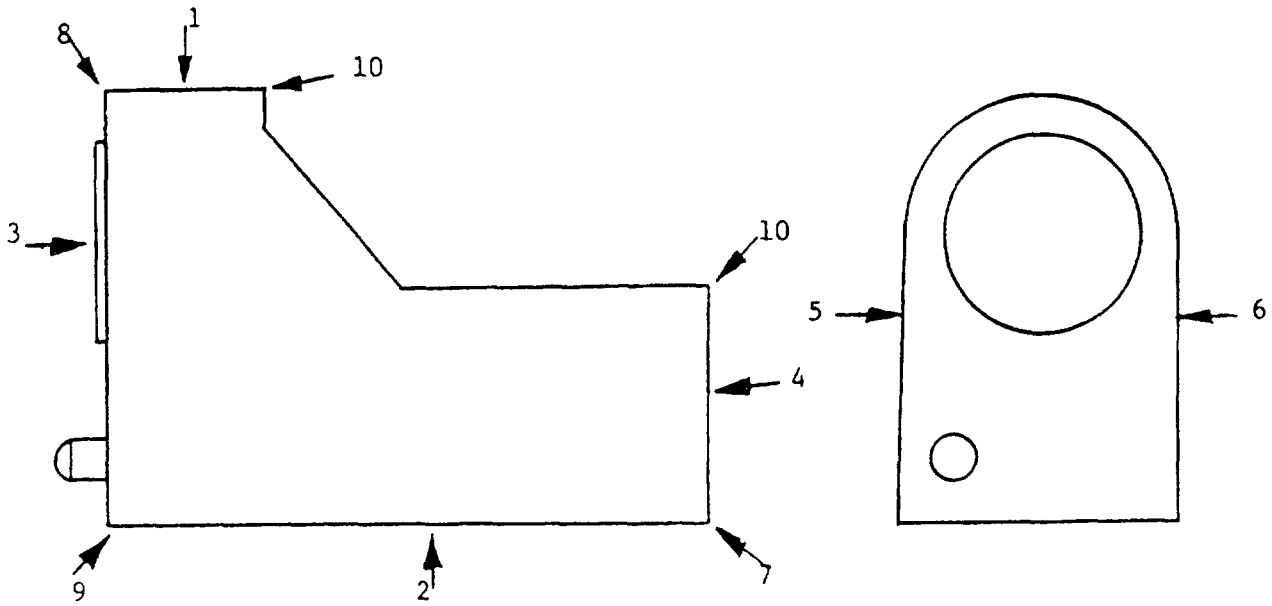


Figure 1. Shock (drop) sequence.

4.6.6.8.4 Fungus. The instrument shall be subjected to the test of Method 508.1, Procedure I of MIL-STD-810.

4.6.6.8.5 Failure criteria. Nonconformance to 3.7 shall constitute failure of these tests.

4.6.7 Bounce preconditioning. After assembly and prior to submission to the Government for acceptance, each instrument shall be rested on its largest face on a bounce machine and subjected to a preconditioning on the bounce table for a minimum period of 1 minute. The bounce table shall be capable of producing a total excursion ot exceeding 0.5 inch (12.7mm) at a rate of 285 $\pm$3 revolutions per minute. Nonconformance to 3.8 shall constitute failure of this test.

4.6.8 Inspection for dimensional interchangeability. The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.9. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of cases, covers and insertable assemblies, when such dimension affect mating of parts.
b. Dimensions of cavities, when such dimensions affect insertion of items.

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c. Location of hinges and fasteners on separable parts of assemblies which must mate, such as cases, covers, and mountings.
d. Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts of plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major unit.
e. Size and form of special threads.

4.6.9 Weight. The radiacmeter, less technical manual, shall be weighed. Nonconformance to 3.10 shall constitute failure of this test.

4.7 Packaging inspection. Packaging inspection requirements specified herein are classified as follows:

a. First article inspection of packaging.
b. Quality conformance inspection of packaging.

4.7.1 First article inspection of packaging. Unless otherwise specified in the contract, first article inspection of packaging shall be in accordance with the unit pack design validation requirements of MIL-P-116.

4.7.2 Quality conformance inspection of packaging.

4.7.2.1 Materials inspection. All materials to be used in packaging shall be inspected in accordance with the applicable material specification.

4.7.2.2 Preservation inspection. Inspection of preservation and interior markings shall be in accordance with group A and B quality conformance inspection requirements of MIL-P-116. Lot formation and sampling procedures shall be as specified therein.

4.7.2.3 Packing inspection. Inspection of packing and the marking for shipment and storage shall consist of the examinations specified in table VI, "Packing inspection provisions". Lot formation shall consist of all packs made of the same materials during an identifiable period and submitted at one time for acceptance. Sampling procedures shall be in accordance with MIL-STD-105, using a single sampling plan and acceptable quality level of 4.0 percent defective.

TABLE VI. Packing inspection provisions.

No.	Characteristic	Method of inspection
101	Intermediate container not as specified.	Visual
102	Improper closure of intermediate container.	Visual
103	Shipping containers not in accordance with specification	Visual
104	Excessive cube.	Visual
105	Improper blocking and bracing.	Visual
106	Closure not in accordance with specification.	Visual
107	Weight and size exceed container limitations.	Weight & measure
108	Strapping not in accordance with specification, incorrectly applied or omitted.	Visual
109	Marking omitted, incorrect, or illegible.	Visual

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5. PACKAGING

5.1 Preservation. Preservation shall be as specified in MIL-STD-726, coded as follows:

5.1.1 Level A. JR-1-1-ØØ-00-NS-X-ED-0-00-A.

5.1.2 Level B. JR-1-1-ØØ-00-NS-X-ED-0-00-B.

5.2 Packing and marking. Packing and marking shall be in accordance with MIL-E-55585.

6. NOTES

6.1 Intended use. Radiacmeter IM-174( )/PD is a portable, survey meter. The meter detects and measures gamma radiation within the intensity range of 1 to 500 rads/hr over the energy range of 80 KEV to 5 MEV. Radiation intensity is indicated on a 2.5 decade pseudo-logarithmic scale.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. When a first article is required for inspection and approval and quantity of first article units (see 3.2).
c. When batteries shall be provided for shipment (see 3.4.2).
d. Group C failure action (see 4.5.3.3).
e. Level A or B preservation and packing (see section 5).
f. When first article inspection rough handling tests are not required.
g. When first article packaging inspection test reports require acquisition activity approval prior to production unit packing.

6.3 First article. When a first article inspection is required, the item will be tested and should be a first article sample. The first article should consist of the quantity of first article units specified in the contract. The contracting officer should include specific instructions in acquisition documents for examination, test, and approval of the first article.

6.4 Government-loaned property. The contracting officer should arrange to loan the property listed in 3.13.

6.5 Nomenclature. The parentheses in the nomenclature will be delted or replaced by a letter identifying the particular design; for example: IM-174W/PD. The contractor should apply for nomenclature in accordance with the applicable clause in the contract (see 1.1).

6.6 Level B preservation. When level B preservation is specified, this level of protection will only be used under known favorable conditions during transportation, storage and handling.

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6.7 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the change.

Custodian:
Army - ER

Preparing activity:
Army - ER
Project 6665-A445

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-55615B(ER)	RADIACMETER IM-174( )/PD

3a. NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording.

b Recommended Wording

c Reason/Rationale for Recommendation

6 REMARKS

7a. NAME OF SUBMITTER *(Last First, MI)* – Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

DD FORM 1426 EDITION OF 1 OCT 75 IS OBSOLETE

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